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Pharmacology of Autacoids and Anti-autacoids: Part I

Histamine, antihistamines and serotonin agonists

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Pharmacology I (PYCH304)

15th Oct 2025 (8.00-10.00)

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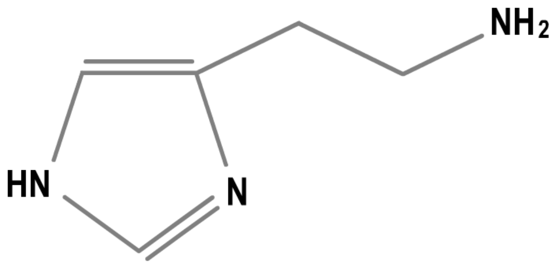
Learning Objectives

After the class, students should be able to

1. Explain mechanism of action and pharmacological effects of H_1 antihistamines
2. Explain pharmacokinetics of H_1 antihistamines
3. Explain adverse drug reactions and drug interactions of H_1 antihistamines
4. Recognize clinical uses of H_1 antihistamines
5. Understand pharmacological properties of serotonin agonists (mainly –triptans)

Introduction to histamine and H_1 antihistamines

- Histamine (β-aminoethylimidazole)
    - Biologically active amine
    - Autacoids



- Histamine has an important role in human health
    - **Allergic inflammation**
    - **Immunomodulatory effects**
    - **Regulator of gastric acid secretion**
    - Modulator of Neurotransmitter release , etc.
- Antihistamine activity was first demonstrated in 1937
- More than 40 H_1-antihistamines are available worldwide

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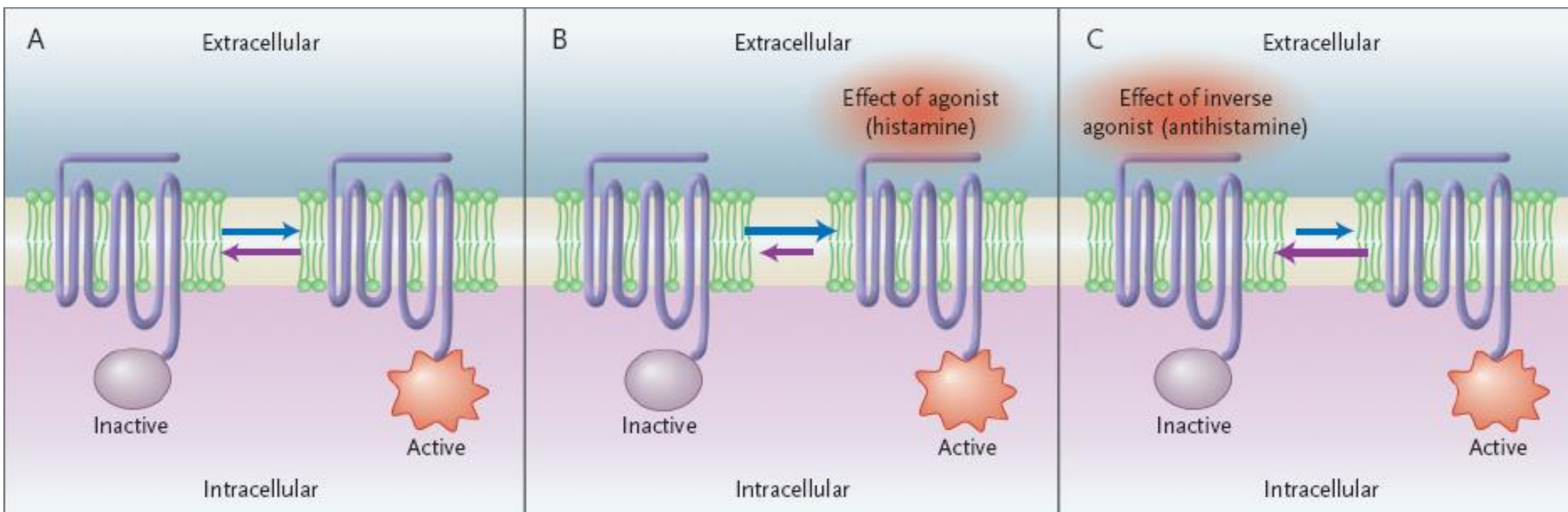
Characteristics of histamine receptor subtypes

Receptor Subtype	Distribution	Post-receptor Mechanism	Partially selective antagonists or Inverse agonists
H1	**Smooth muscle, endothelium, brain**	**Gq, ↑ IP3, DAG**	**Mepyramine,[1] triprolidine, cetirizine**
H2	Gastric mucosa, cardiac muscle, mast cells, brain	Gs, ↑ cAMP	Cimetidine,[1] ranitidine[1]
H3	Presynaptic autoreceptors and heteroreceptors: brain, myenteric plexus, other neurons	Gi, ↓ cAMP	Tiprolisant,[1] Pitolisant[1] **Betahistine[1]**
H4	Eosinophils, neutrophils, CD4 T cells	Gi, ↓ cAMP	Thioperamide[1]

[1]Inverse agonist.
cAMP, cyclic adenosine monophosphate; DAG, diacylglycerol; IP3, inositol trisphosphate.

Mechanism of action of H_1 antihistamines

- Reversible competitive inhibitors at histamine H_1 receptor
- Have been reclassified as inverse agonists
  - Even in the absence of ligand binding, H_1 receptor can trigger downstream events itself
  - **Inverse agonist has a preferential affinity for the inactive state of H_1 receptor and stabilizes the receptors in this conformation**



N Engl J Med 2004;351:2203-2217.

Classification of H_1 antihistamines

Chemical Class	Functional Class	
	1st Generation (Sedating)	2nd Generation (Non-sedating)
Alkylamines	**Chlorpheniramine**, **Brompheniramine**, Dimethindene, Triprolidine, Pheniramine	Acrivastine
Piperazines	**Hydroxyzine**, Chlorcyclizine, Oxatomide, Buclizine, Cyclizine, Meclizine	**Cetirizine**, **Levocetirizine**
Piperidines	**Cyproheptadine**, **Ketotifen**, Azatadine, Diphenylpyraline	**Loratadine**, **Desloratadine**, **Rupatadine**, **Fexofenadine**, **Bilastine**, **Olopatadine**[b], Terfenadine*, Astemizole*, Ebastine, Levocabastine, Mizolastine
Ethanolamines	Carbinoxamine, **Diphenhydramine**, **Dimenhydrinate**, Doxylamine, Phenyltoloxamine, Clemastine	
Ethylenediamines	**Antazoline**[b], Pyrilamine, Tripelenamine	
Phenothiazines	**Promethazine**, Methdilazine	
Others	**Doxepin**[a], **Mebhydrolin**	Azelastine, Emedastine, Epinastine

May be classified as 3rd Gen

* Has had approval withdrawn ; [a] Has H_1 and H_2 antagonists and is also classified as a Tricyclic antidepressant ; [b] Topical H_1 antihistamine is applied to the conjunctivae

J Allergy Clin Immunol. 2011;128(6):1139-1150.e4.

Pharmacological properties of H_1 antihistamines

- Most H_1 antagonists have similar pharmacological actions and therapeutic applications
  - Smooth Muscle
  - Capillary permeability
  - Exocrine Glands
  - Flare and Itch
  - Immediate Hypersensitivity Reactions
  - Local Anesthetic Effect

Effect of H_1 antihistamines on respiratory smooth muscle

Histamine Ξ H_1 receptor in Respiratory smooth muscle

Antihistamines

Bronchoconstriction

Bronchoconstiction is reduced little

Coz' this includes other mediators

e.g. Leukotrienes , PAF

Effect of H_1 antihistamines on vascular tree

Histamine Ξ H_1 receptor at vascular endothelial cell

Stimulate ↓

eNOS

↑NO

↓

Vasodilation

Nasal arterioles (Nasal congestion)

Systemic vessels (Hypotension, Flushing, Headache)

Eye arterioles (Red eye)

Antihistamines

↓

Inhibit vasodilator effect of histamine

- ↓ Nasal congestion, Red eye
- ↓ Histamine-induced changes in systemic blood pressure

Effect of H_1 antihistamines on capillary permeability

Histamine Ξ H_1 receptor at post-capillary venules

Endothelial cell contract and separate at their boundaries

↑ Plasma protein and Fluid permeability (Extravasation)

Skin capillaries (Edema , Weal)

Nasal capillaries (Rhinorrhea)

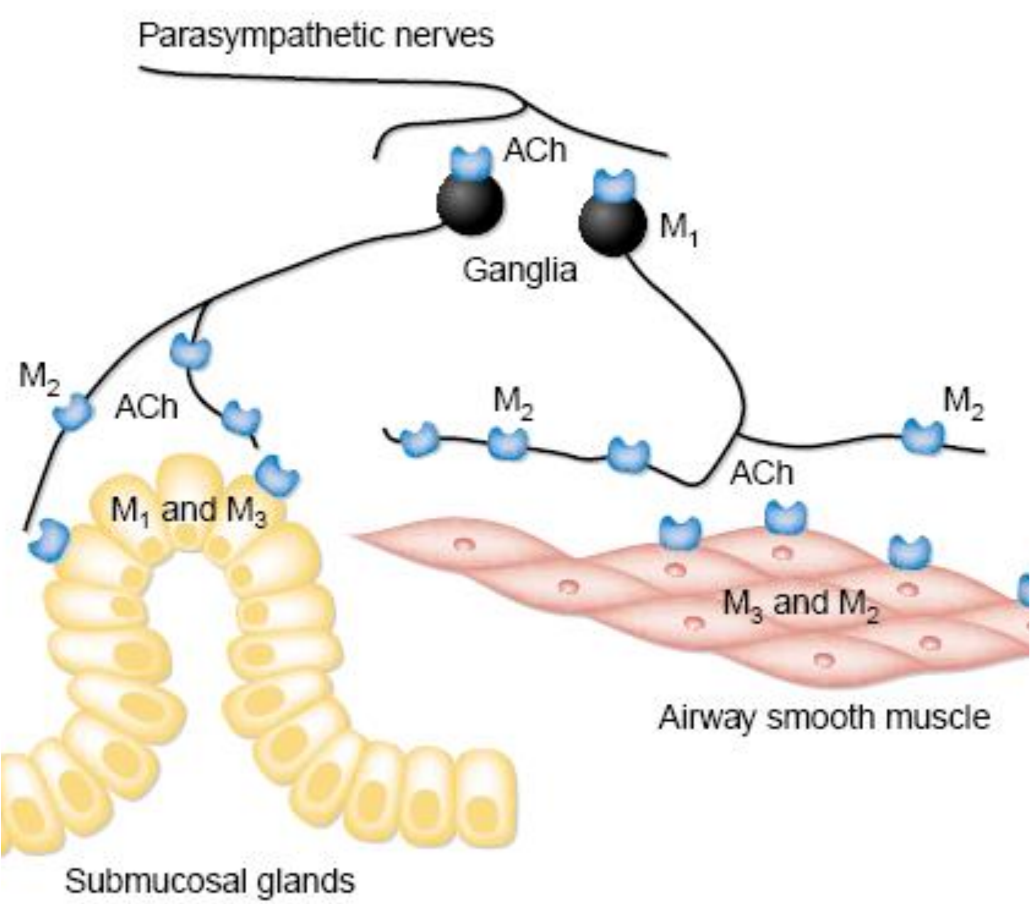
Antihistamines

Strongly block capillary permeability

↓ Edema , Weal
↓ Nasal discharge

Effect of H_1 antihistamines on exocrine glands

- Histamines stimulate parasympathetic activity → Rhinorrhea



Curr Opin Pharmacol 2001, 1:223–229

- 1st generation drugs have anticholinergic effect
  - May tend to lessen Rhinorrhea
- 2nd generation drugs have no effect on Muscarinic receptors
  - Cetirizine has minimal anticholinergic effect

Effect of H_1 antihistamines in flare and itch

Histamine Ξ H_1 receptor at Sensory nerve endings

SKIN

- Epidermis → Itch
- Dermis → Pain and Itching
- Stimulate axon reflexes → Indirect Vasodilation → Flare

Eyes, Ears, Nose, Palate, Throat → Itching

Antihistamines

Other pharmacological properties

- Immediate hypersensitivity reactions
  - Histamine is one of the many contributing autacoids
  - Effects of H_1 antihistamines
    - Edema formation and itch are effectively suppressed
    - Other effects are less well antagonized
      - Hypotension
      - Bronchoconstriction
- Local anesthetic effect of H_1 antihistamines
  - Promethazine is especially active
  - Require higher concentration than those which antagonize H_1 receptor

Pharmacokinetics of H_1 antihistamines

- Absorption : well absorbed from GI tract
- Onset : 15-30 mins
- T_{max} : 1-3 hrs
- Duration of action
  - 1st generation are usually last 4-6 hrs
  - 2nd generation are much longer acting (12-24 hrs)
- Distribution
  - Widely throughout the body
  - Peak conc. achieved rapidly in the skin and persist (≥ 36 hrs) even when plasma level are very low

Pharmacokinetics of H_1 antihistamines (Cont.1)

- Metabolism
  - 1^{st} generation are metabolized by CYPs
  - 2^{nd} generation
    - Loratadine $\xrightarrow[\text{CYP2D6}]{\text{CYP3A4}}$ Descarboethoxyloratadine (Desloratadine)

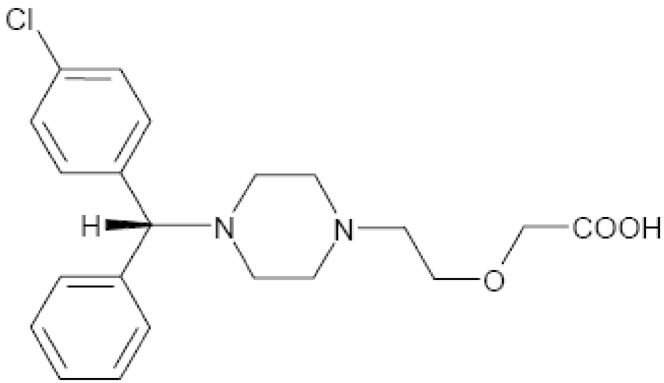
      Higher potency
    - Terfenadine $\xrightarrow{\text{CYP3A4}}$ Fexofenadine
    - **Minimal/No hepatic metabolisms**
      - **Cetirizine** (active metabolite of hydroxyzine)
      - **Fexofenadine** (active metabolite of terfenadine)
      - **Bilastine**

Int J Mol Sci. 2019;20:213.

Pharmacokinetics of H_1 antihistamines (Cont.2)

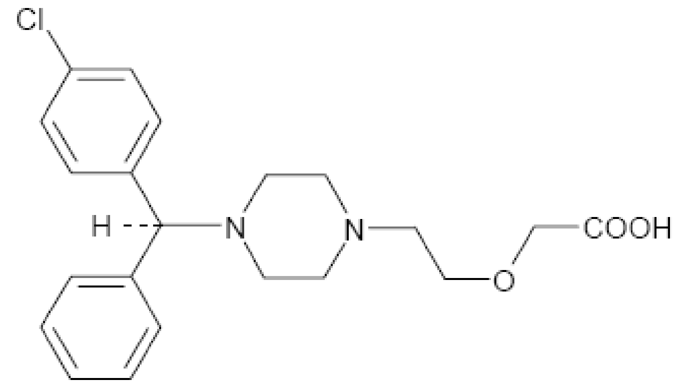
- Excretion
  - 1st generation are excreted mainly in the urine as metabolites
  - 2nd generation
    - Cetirizine, loratadine, fexofenadine are excreted mainly in the unmetabolized form
      - Cetirizine, loratadine are excreted primarily into the urine
      - Fexofenadine is excreted primarily in the feces

Enantioselectivity of Cetirizine



(*R*)-levocetirizine

(the eutomer)



(*S*)- dextrocetirizine

(the distomer)

Biochem Pharmacol 2003;66:1123–1126

- In binding assays, levocetirizine has demonstrated
  - 2-fold higher affinity for the H_1 receptor compared to cetirizine
  - Approximately 30-fold higher affinity than dextrocetirizine
  - Longer dissociation half-life than dextrocetirizine

Adverse Effects of H_1 antihistamines

- **Central Nervous System**
  - Stimulation
  - Depression
- **Anticholinergic effects**
- **Cardiovascular system**
  - Postural hypotension
- **Others**

CNS side effects of H_1 antihistamines

CNS Stimulation

- Occasionally in Infants and young children "*Paradoxical excitation*"
- Alkylamines (e.g. Chlorpheniramine) are more common
- Normal doses
  - Restless
  - Nervous, Irritability
  - Hyperactivity
  - Insomnia
- Overdose
  - Convulsion (particularly in infants)

CNS Depression

- Dizziness
- ↓ Alertness

Histamine Ξ H1 receptor in CNS

Wakefulness

Cognition

Antihistamines

Sedation , somnolence , hangover in the morning (Impaired cognitive and psychomotor performance with or without perceptible sedation)

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CNS side effects of H_1 antihistamines (Cont.)

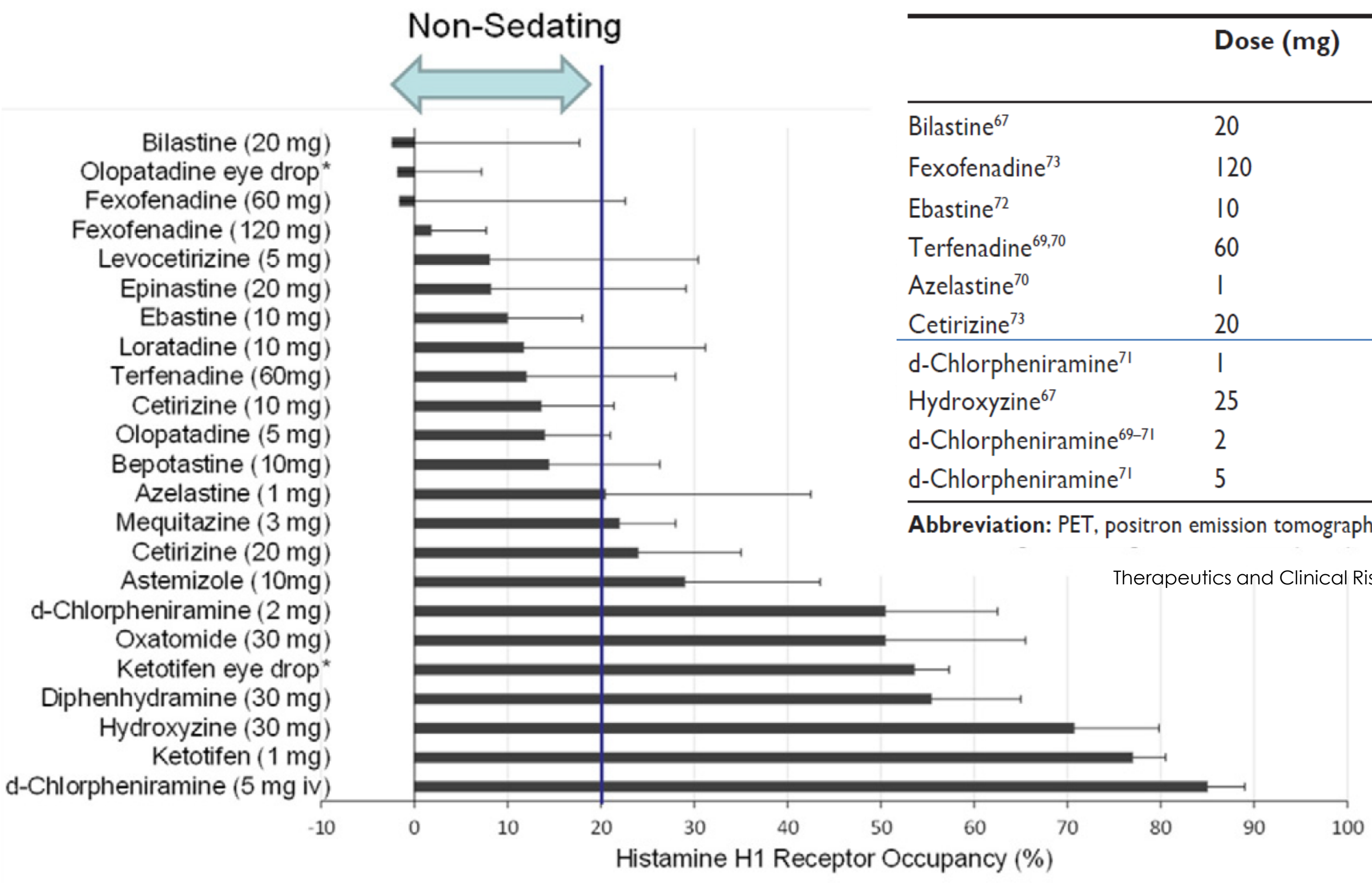
1st Generation Drugs

- Both stimulate and depress CNS
- Lipophilicity
- Largely cross the BBB
- All can cause sedation
  - **Strong**
    - Diphenhydramine
    - Dimenhydrinate
    - Promethazine
    - Hydroxyzine
    - Doxepin
  - **Less potent than others**
    - Alkylamines e.g. CPM

2nd Generation Drugs

- Lipophobicity (and are ionized at physiologic pH)
- Poorly penetrate the BBB
- Affinity for P-gp efflux pump
- Less sedative effects
  - Cetirizine has mild sedating effect **(Higher incidence of drowsiness than others)**
    - Active metabolite of hydroxyzine
    - 30 percent propensity to occupy H_1 receptor in CNS

Comparative sedating effect of antihistamines



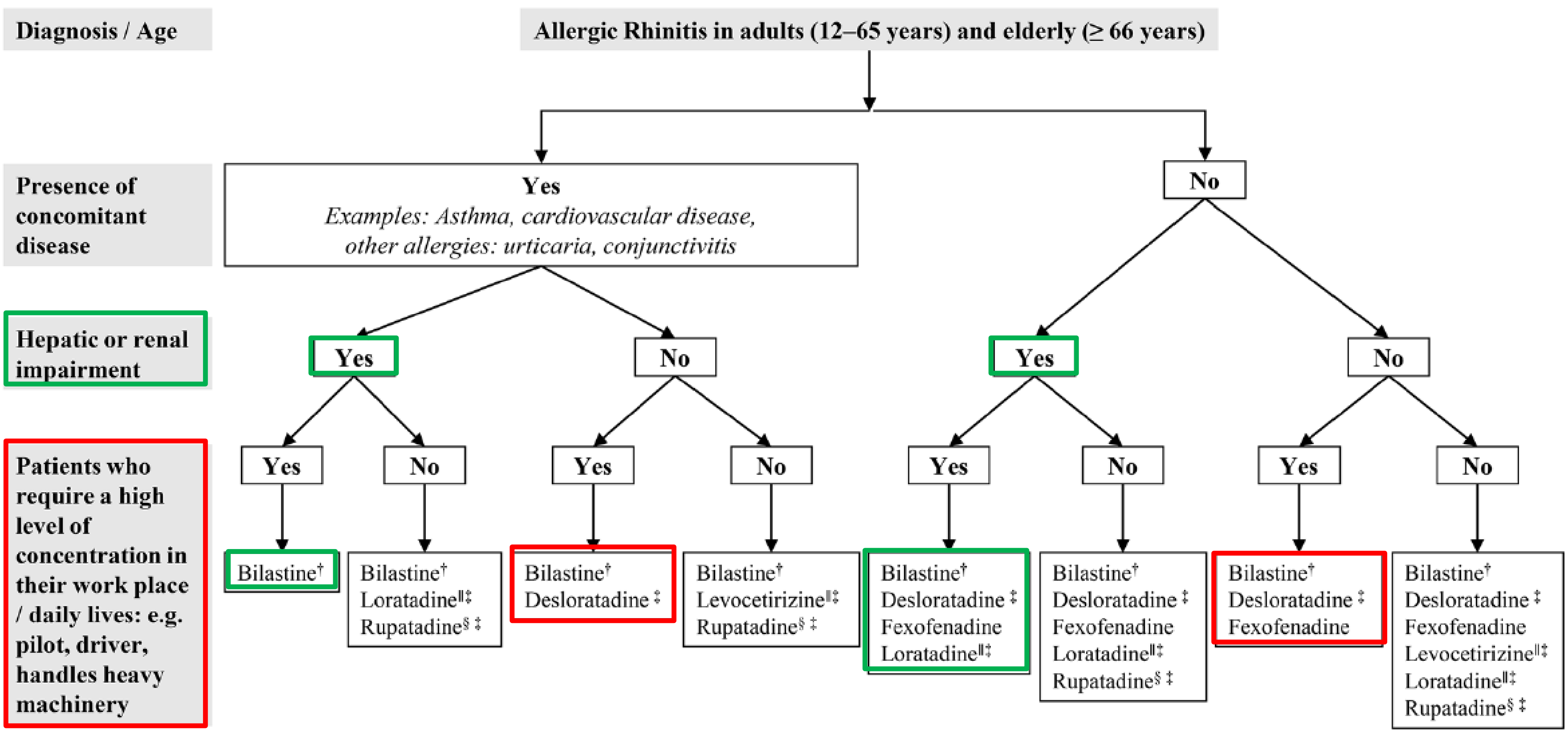
Pharmacol Ther. 2017;178:148-156.

	Dose (mg)	Mean H_1 receptor occupancy on PET (%)
Bilastine[67]	20	−3.92
Fexofenadine[73]	120	−0.1
Ebastine[72]	10	9.9–14.4
Terfenadine[69,70]	60	12.1–17.2
Azelastine[70]	1	20.3
Cetirizine[73]	20	26.0
d-Chlorpheniramine[71]	1	40.4
Hydroxyzine[67]	25	53.95
d-Chlorpheniramine[69–71]	2	60.4–76.8
d-Chlorpheniramine[71]	5	85.5

Abbreviation: PET, positron emission tomography.

Therapeutics and Clinical Risk Management 2016:12;585–97.

An example of antihistamine of choice in patients with underlying clinical conditions



Clin Mol Allergy. 2017;15:19.

Anticholinergic side effects of H_1 antihistamines

- H_1 receptors have approximately 45 percent homology with Muscarinic receptors
- Many 1st generation tend to inhibit Muscarinic receptors
  - **Potent**
    - Promethazine (Strongest)
    - **Diphenhydramine (Clinical use)**
    - Dimenhydrinate (Diphenhydramine + 8-Chlorotheophylline)
    - Doxepin
- 2nd generation have no effect on Muscarinic receptors
  - Cetirizine has minimal anticholinergic effect

Anticholinergic side effects of H_1 antihistamines (Cont.)

- **More common**
  - Dry mouth
  - Dry eye, Mydriasis , Blurred vision
  - Urinary retention
  - Constipation
- **Less Common**
  - Memory deficit

Other adverse effects of H_1 antihistamines

- Postural hypotension
  - α adrenergic receptor blocking action
  - Especially those in the Phenothiazines
    - Promethazine
- Rare
  - Weight gain
  - Drug allergy (Commonly from topical application)
  - Allergic dermatitis

Pregnancy and Lactation

- Pregnancy category B
  - 1st generation : Chlorpheniramine, Diphenhydramine
  - 2nd generation : Cetirizine, Loratadine
- Lactation
  - Antihistamines can be excreted in small amounts in breast milk
  - 1st generation taken by lactating mothers may cause symptoms in the nursing infant
    - Irritability
    - Drowsiness
    - Respiratory depression

Precautions and Contraindications

- Use with caution in
  - Pregnant women
  - Narrow-angle glaucoma
  - Symptomatic prostatic hypertrophy

Anticholinergic effect

- Patient should avoid activities requiring mental alertness until drug effects are realized
  - Drivers
  - Pilots
- Contraindications
  - Hypersensitivity to Antihistamines

Clinical Uses of H_1 antihistamines

- Allergic Diseases
- Common cold
- Motion sickness, vertigo
- Appetite stimulation
  - Cyproheptadine has antiserotonin activity (↑ Appetite)
- Management of Parkinsonism (EPS)
  - Diphenhydramine has potent anticholinergic activity and can be used in management of Antipsychotic-induced Extrapyramidal symptom (EPS)

Allergic Diseases

- Allergic rhinitis
  - Rhinorrhea, Nasal itching, Sneezing, Nasal congestion
- Allergic conjunctivitis
  - Red eye, Eye itching
- Urticaria
- Other allergic reactions
  - Itching of Palate, Throat, Ears
  - Insect bite and stings

J Eur Acad Dermatol Venereol. 2015;29 Suppl 3:16-32.

- Drug allergy (Not anaphylaxis and angioedema)
  - Adjuvant therapy of systemic anaphylaxis and angioedema

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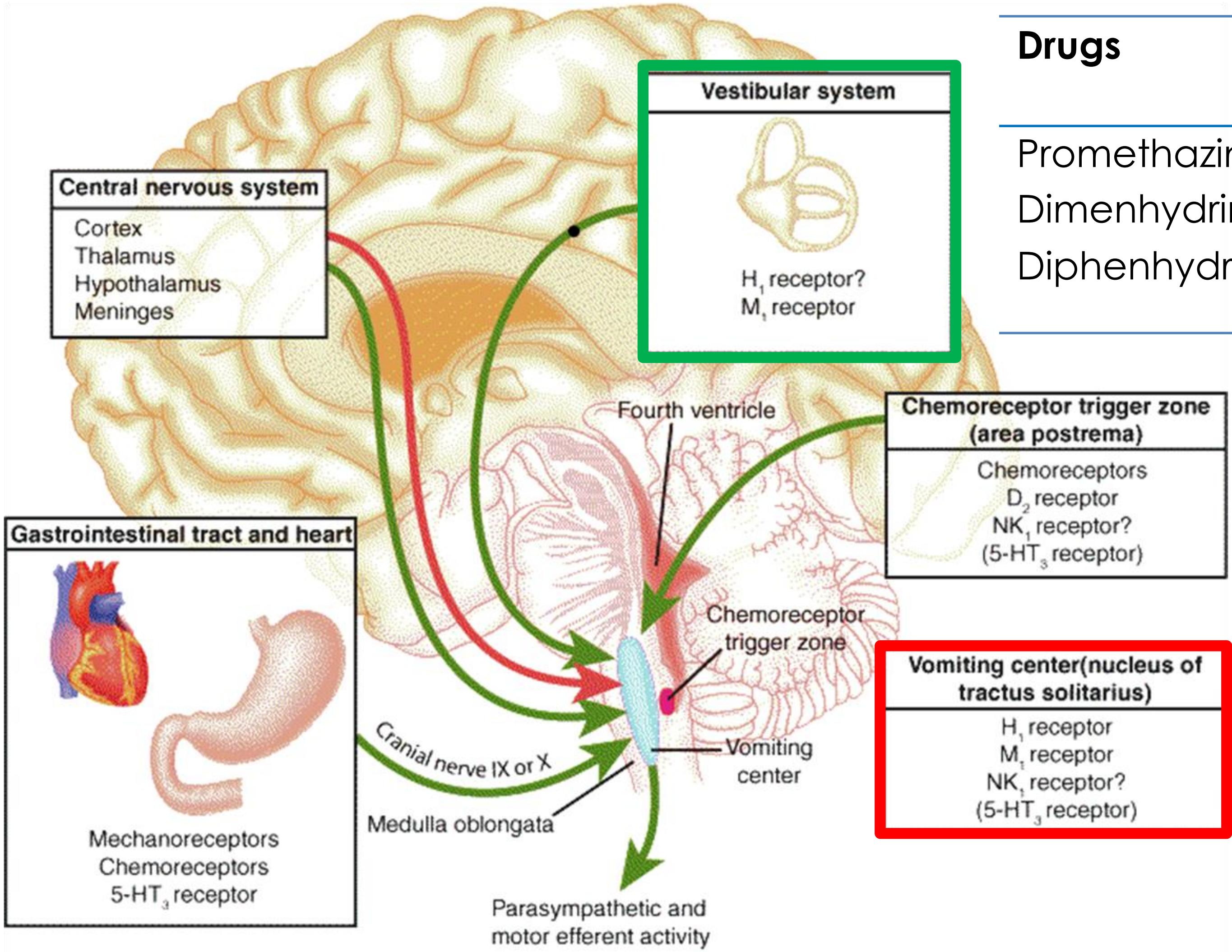
Allergic Diseases (Cont.)

N Engl J Med 2002;346:175-179.

- Allergic dermatoses respond favorably to H_1 antihistamines
- **Hydroxyzine and Cetirizine**
  - Enter the skin readily, and sustained high conc. in skin
  - Well-known efficacy in symptomatic treatment of urticaria and other skin disorders
- **Chronic urticaria**
  - **May need high dose, particularly 2nd gen.**
  - **Combined use of H_1 and H_2 antagonists sometimes is effective**
  - **Doxepin** is about 800 times more potent antihistamine activity than Diphenhydramine and can be effective

Allergy. 2018;73:1393–1414.

Motion Sickness, Vertigo



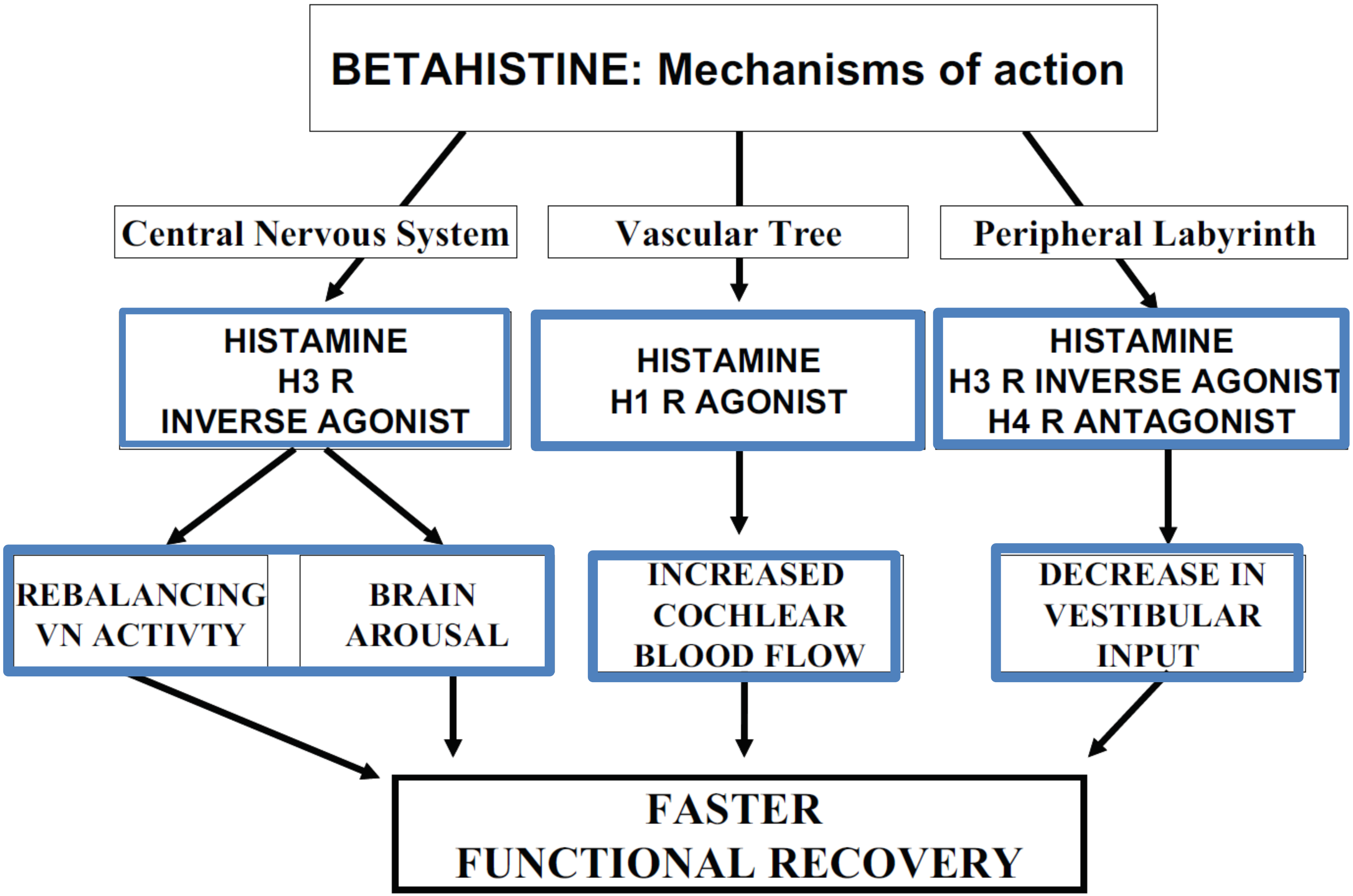
Source: Katzung BG, Masters SB, Trevor AJ: *Basic & Clinical Pharmacology, 11th Edition*: http://www.accessmedicine.com

Drugs	**Proposed sites of action**
Promethazine	D_2, H_1, $\mathbf{M_1}$
Dimenhydrinate	H_1, $\mathbf{M_1}$
Diphenhydramine	H_1, $\mathbf{M_1}$

J Perianesth Nurs 2006;21:385-397

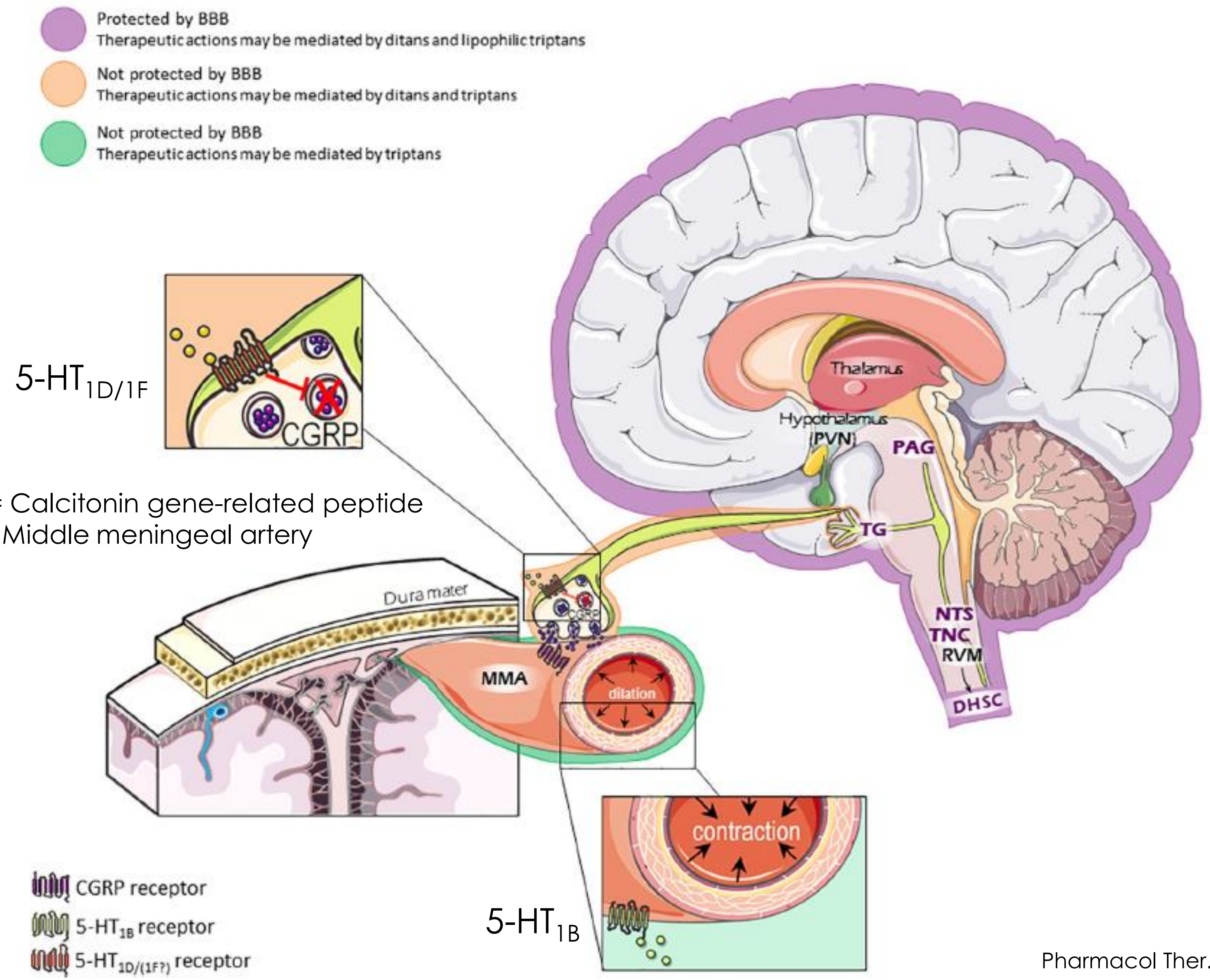
The anti–motion sickness effects are rather due to their central-acting anticholinergics

Betahistine



J Vestib Res. 2013;23(3):139-51.

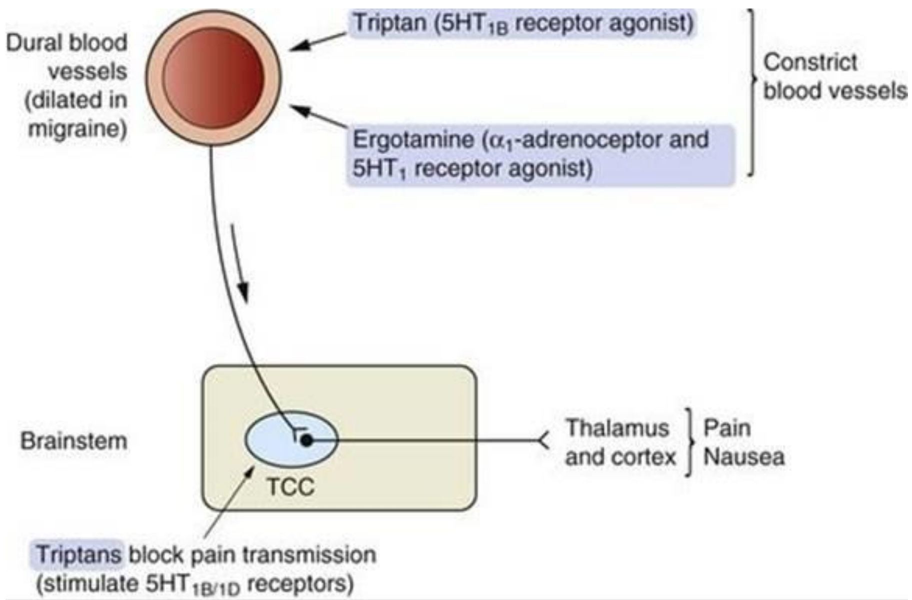
The role of serotonin (5-HT) in migraine headache

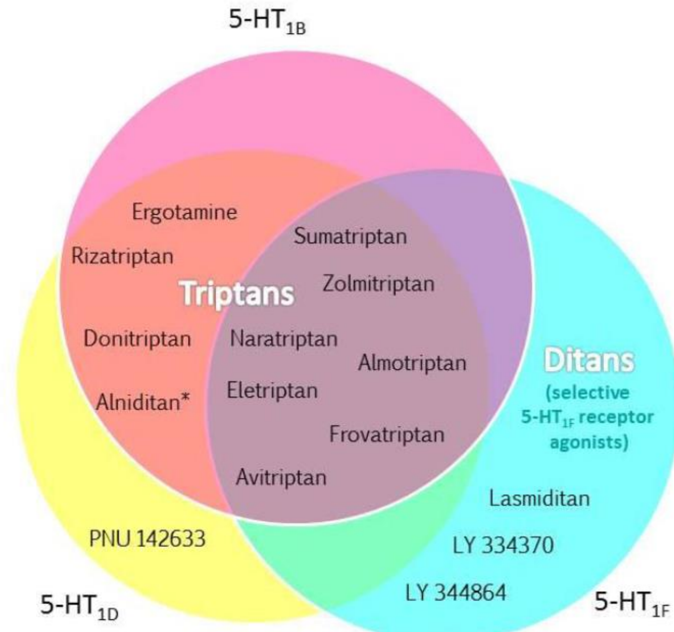


CGRP = Calcitonin gene-related peptide
MMA = Middle meningeal artery

Pharmacol Ther. 2018;186:88-97.

Pharmacology of 5-$HT_{1B/1D}$ agonists (Triptans)





Pharmacol Ther. 2018;186:88-97.

Variable	Sumatriptan	Almotriptan	Eletriptan	Frovatriptan	Naratriptan	Rizatriptan	Zolmitriptan
Half-life (hr)	2.0	3.5	5.0	25.0	5.0–6.3	2.0	3.0
Time to maximal concentration (hr)							
During attacks	2.5	2.0–3.0	2.8	3.0	—	1.0	4.0
At other times	2.0	1.4–3.8	1.4–1.8	3.0	2.0–3.0	1.0	1.8–2.5
Oral bioavailability (%)	14	69	50	24–30	63–74	40	40
Metabolism and excretion							
Primary route	MAO-A	CYP450 and MAO-A	CYP3A4	Renal, 50%	Renal, 70%	MAO-A	CYP450-1A2
Secondary route	—	—	—	—	CYP450	—	MAO-A

Active met.

*Data are derived from multiple studies.[73-86] MAO denotes monoamine oxidase, CYP450 cytochrome P450, and CYP3A4 the 3A4 isoform of cytochrome P450.

[1]N Engl J Med. 2002;346(4):257-70.[2]CNS Drugs. 2012;26:949–957.

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ADRs and drug interactions of ergotamine and triptans

- **ADRs**
  - Hypertension (5-HT$_{1B}$ receptor, and a$_1$-receptor in case of ergots)
  - Coronary vasoconstriction (5-HT$_{1B}$ receptor)
    - **Contraindication in patients with cardiovascular risk factors**
  - **Teratogenic (Ergotamine)**
- **Drug interactions**
  - Ergotamine and CYP3A4 inhibitors
    - e.g. Protease inhibitors (anti-HIV drugs)
    - **Contraindication for concomitant use**
  - Triptans and MAO-A inhibitors

Acute ischemia of the limb, leading to gangrene

Circulation. 2011;124:1395-1397.

Thank You

Any Questions?